

Program : Diploma in Chemical Engineering	
Course Code : 5078	Course Title: Technical Analysis Lab
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To impart basic knowledge on the analysis of consumable chemicals

Course Pre-requisites:

Topic	Course code	Course name	Semester
Oils, fats, soaps and bleaching agents.		Organic Chemical Technology	4

Course Outcomes :

On completion of the course student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Analyze the basic characteristics of oils.	15	Applying
CO2	Analyze the basic characteristics of soap.	18	Applying
CO3	Analyse sugar	6	Applying
CO4	Determine the quality of bleaching powder	6	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Analyze the basic characteristics of oils.		
M1.01	Estimate the acid value of oil	3	Applying
M1.02	Estimate the Iodine value of oil	3	Applying
M1.03	Estimate the saponification value of oil	3	Applying
M1.04	Open ended experiments	6	Applying
CO2	Analyze the basic characteristics of soap.		
M2.01	Estimate the moisture content of soap	3	Applying
M2.02	Estimate the total alkali content of soap	3	Applying
M2.03	Estimate the alcoholic insoluble's in soap	3	Applying
M2.04	Estimate the water insoluble's in soap	3	Applying
M2.05	Estimate the total fatty matter in soap	3	Applying
M2.06	Open ended experiments	3	Applying
	Series Test – I		
CO3	Analyse sugar		
M3.01	Estimate the sucrose content in sugar	3	Applying
M3.02	Open ended experiments	3	Applying
CO4	Determine the quality of bleaching powder		
M4.01	Estimate the available chlorine content in the bleaching powder.	3	Applying
M4.02	Open ended experiments	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Engineering Chemistry; Jain and Jain
R1	ASTM D820 Standard test methods for the analysis of soaps.

Online Resources

Sl.No	Website Link
1	https://www.vlab.co.in/broad-area-chemical-engineering